

Nouredine Khaous

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PhD student in Natural Language and Speech Processing, specializing in Automatic Speech Recognition (ASR). My research focuses on improving speech models in multilingual and low-resource settings by understanding how they process and represent linguistic information, and using this knowledge to build better and more robust models.

WORK EXPERIENCE

- **Teaching Assistant** Sep 2026 – Present
Université Paris Cité Paris, France
 - Teaching Introduction to Python, covering programming fundamentals and supervising hands-on practical sessions.
- **Data Scientist** Sep 2023 – Sep 2025
Orange [🌐] Paris, France
 - Adapted large language models to Wolof through dataset construction, vocabulary expansion, and distributed fine-tuning with DeepSpeed, achieving a 12% METEOR improvement on Wolof→French and Wolof→English machine translation.
 - Built and deployed an Agentic RAG system with LangGraph for network incident management, integrating summarization, chatbot support, and next-action prediction, reducing ticket processing time by 50%.
- **Research Intern** May 2025 – Jul 2025
Bergische Universität Wuppertal Remote
 - Collaborating on AI models integrating physics for fluid state prediction in nonlinear port-Hamiltonian systems.
 - Developing conditional diffusion models to generate spatially-resolved system state predictions in image form.
 - Supporting system identification and simulation of complex fluid dynamics processes.
 - Conducting research at the intersection of machine learning, physics-informed AI, and dynamical systems.
- **Research Intern** Oct 2024 – Jan 2025
IDEMIA [🌐] Paris, France
 - Improved contactless fingerprint image quality using state-of-the-art diffusion models for biometric applications.
 - Designed an attention-based U-Net architecture with time-step encoding for image denoising.
 - Enhanced similarity between contactless and contact-based fingerprint images, improving recognition robustness.
 - Conducted experiments on generative image restoration methods for secure identity systems.

EDUCATION

- **PhD in Machine Learning for Low-Resource Languages** Feb 2026 – Present
Université Paris Cité | Laboratoire de Linguistique Formelle Paris, France
 - Topic: Machine learning for rare languages: linking speech recognition and analysis of the internal representations of Speech Models in service of field linguistics | Supervisor: [Guillaume Wisniewski](#)
- **Master of Science in Computer Science and Robotics** Sep 2023 – Sep 2025
Sorbonne Université Paris, France
 - Focus on Machine Learning, Speech Processing, Computer Vision and Robotics.
 - Dissertation Topic: Adapting Large Language Models to Low-Resource Languages: A Case Study of Wolof.
 - Graduated Rank: 3/45
- **Bachelor of Engineering in Electronics and Control Theory** Sep 2021 – May 2023
Université Claude Bernard Lyon 1 Lyon, France
 - Focus on Electronics, Electrical Engineering, and Control Theory.
 - Graduated Rank: 4/50
- **Preparatory Classes for Engineering Schools | Mathematics and Physics Track** Sep 2019 – Jun 2021
École Nationale Polytechnique Algiers, Algeria
 - Intensive coursework in Mathematics, Linear Algebra, Physics, Chemistry, and Engineering Sciences.
 - Graduated Rank: 19/283

PUBLICATIONS

- **Khaous, N.** & Wisniewski, G. (2025). *Accent-Induced Geometric Displacement in Speech Representations Predicts Automatic Speech Recognition Degradation*. [TALN 2026](#).
- El Messaoudi, A., **Khaous, N.**, Cherifi, K. (2026). *Target Parameterization in Diffusion Models for Nonlinear Spatiotemporal System Identification*. [arXiv preprint 2604.17566](#).

SKILLS

- **Programming Languages:** Python, C/C++, SQL, Bash
- **Machine Learning & AI:** PyTorch, Hugging Face, Deep Learning, NLP, LLMs, Transformers, Computer Vision
- **Data Science & Scientific Computing:** Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn
- **Tools & Technologies:** Docker, Git, Linux, OpenCV, LaTeX
- **Research Areas:** Speech Processing, Representation Learning, Low-Resource Languages, Multilingual NLP
- **Languages:** French (Native), Arabic (Native), Kabyle (Native), English (Full Professional Proficiency)